
INTERFACE CONTROL DOCUMENT

This document describes mechanical, optical, electrical and software interface of the following Noptel LRF family modules:

- **LRF126-M4**
- **LRF127-M4**



Figure 1. LRF126-M4 and LRF127-M4

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Document version:

Version	Date	Author	Firmware	Notes
2.0	11.5.2020	Juha Leskinen	1.4.22	All measurement commands replaced by new general measurement command (CCh). Previous measurement commands are still supported (doc. OS43830CE)
2.0	11.5.2020	Matti Hiekkaranta		Chapters reorganization. Added info of transmitter divergence.
2.1	29.6.2020			Frame bits corrected.
2.2	10.12.2020	Juha Leskinen	1.5.3.	New electronics version & FW. Values updated accordingly.
3.0	01.12.2021	Jani Mäklin	X.5.5	Separated LRF126 and LRF127 from the document OS43830.
3.1	27.01.2022	Jani Mäklin		Updated the performance table. Reviewed and released.
3.2	15.5.2023	Kai Väisänen	2.6.x. ->	New O50052BE version. Added 500 Hz CMM measuring mode. Added faster baud rates. Typical range performance table updated.
3.3	25.6.2024	Kai Väisänen	2.6.x. ->	Changed mechanical interface document numbers and updated weight information.

1 General

This chapter includes references for mechanical interfaces of each module and information about eye-safety and optical crosstalk.

1.1 Mechanical interface

Dimensions, fixing points and mechanical alignment of the LRX's are shown in separate documents as follows:

Table 1: Mechanical interface document and dimensions of each module

Module	Document number	Dimensions [mm]	Weight [g]
LRF126-M4	Y100111	48 x 34 x 20	34
LRF127-M4	Y100093	48 x 34 x 20	36

1.2 Eye-safety

The LRF range finder modules are classified to Class 1. The LRF is eye-safe in every measurement mode.

The system which uses the LRF module shall have labelling according to IEC 60825-1:2014 as follows:

Text borders and symbols are recommended to be black on a yellow background, but for Class 1 this color combination need not be used.

The labels shall be durable, permanently affixed, legible, and clearly visible during operation, maintenance, or service, according to their purpose.

If the size or design of the product makes labelling impractical, the label shall be included with the user information or on the package.

Direct printing or engraving of equivalent labels on the laser product or panels is acceptable.

Class 1 laser product shall have affixed an explanatory label bearing the words:

CLASS 1 LASER PRODUCT

Alternatively, the label illustrated in Figure below may be affixed to the product:

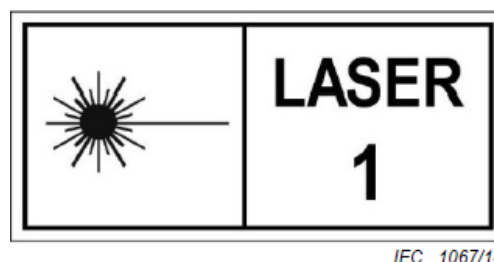


Figure 2. Alternative label for Class 1

1.3 ESD attention

LRF modules are open electronic devices. Handle modules only in ESD safe environment to avoid damaging the devices.



Figure 3. ESD attention.

1.4 Transmitter divergence

The following table represents transmitter divergence of each LRF module.

Table 2: Horizontal and vertical transmitter divergences

LRF Module	LRF126-M4	LRF127-M4
Horizontal Divergence [mrad]	0.5	0.5
Vertical Divergence [mrad]	2.5	2.5

The following pictures illustrates principal transmitter beam divergence and orientation of different LRF modules (reference doc.: T44218A).

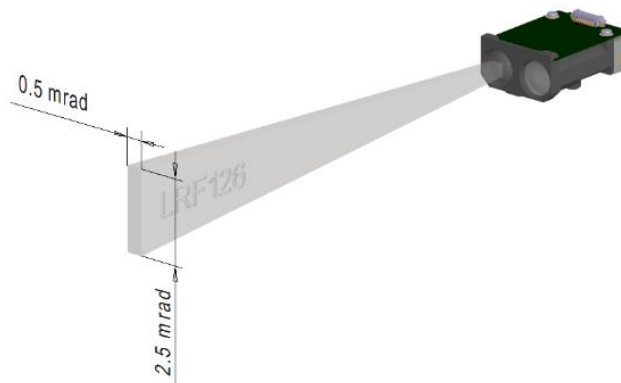


Figure 3. Transmitter divergence of LRF126 and LRF127.

1.5 Optical crosstalk

When installing the LRF module in the user's housing it is important to minimize the level of optical crosstalk between transmitter and receiver.

Level of optical crosstalk can be checked as described in the chapter "3.3 Check optical crosstalk".

Front glass requirements:

- Normal BK7 clear glass operates well with anti-reflection coating at 1.55 μm . Thickness is not critical, but optical crosstalk can get worse with very thick glass; ≤ 3 mm is fine.
- The glass should be installed parallel to front surface and **as close to LRF as possible**. Possible gap should be filled with O-ring type solution or another method to prevent optical crosstalk between the receiver and transmitter optics.

Take also care that mechanical structures in front of LRF are not reflecting laser radiation back to the LRF. Matte black paint can be used to reduce reflections.

1.6 Disposal



Do not dispose this product as unsorted municipal waste.



In the European Union, all electrical and electronic products, batteries, and accumulators must be taken to separate collection at the end of their working life.

2 Electrical interface

This chapter describes electrical interface including connector pinout and typical supply power consumption details.

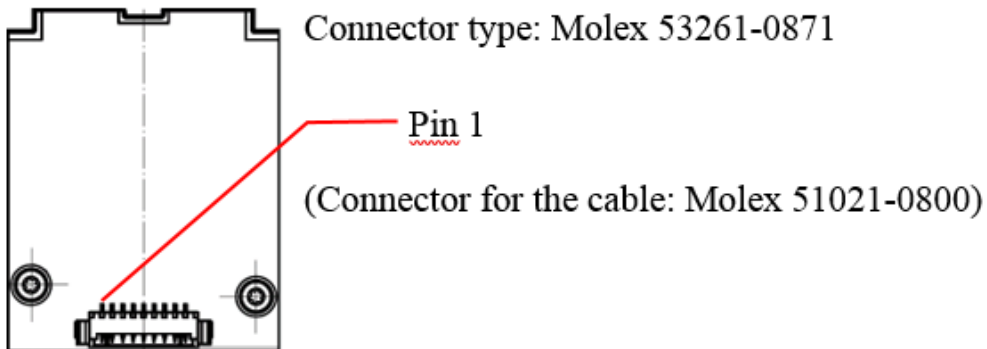


Table 3: Molex connector pinout

pin #	IN/OUT	Description
1	IN	UART RXD (serial data receive, 3.3 V)
2	OUT	UART TXD (serial data transmit, 3.3 V)
3	IN	POWER ON (pull up >1.4V, internal 510kOhm pull-up resistor) SHUT DOWN (pull down <0.4V) *1
4	IN	Supply voltage input (3.3 – 5.4 V) *2 Undervoltage and overvoltage protected.
5	IN	Supply voltage input (3.3 – 5.4 V) *2 Undervoltage and overvoltage protected.
6	GND	Ground *2
7	GND	Ground *2
8	I/O	Reserved signal for optional features

*1. Power control signal can be left floating when not used. Open collector logic recommended for controlling. Inputs 3, 4 and 5 can be shorted.

*2 Connect all supply and ground wires. Use short wires to reduce losses. Recommended supply range 4.6-5.4V.

Table 4: Typical power consumption in different measurement modes.

Module	LRF126/127-M4		Common
Measurement mode (See chapter 3.2)	Current [A _{avg}] *1	Power [W _{avg}]	In-rush Current [A]*2
SMM	0.45	1.8	1.2
QSMM1 & QSMM2	0.45	1.8	1.0

Module	LRF126/127-M4		Common
Measurement mode (See chapter 3.2)	Current [A _{avg}] *1	Power [W _{avg}]	In-rush Current [A]*2
CMM 1 Hz – 200 Hz	0.30	1.2	1.0
CMM 500 Hz	0.2	0.8	1.0
Pointer enabled	0.2	0.8	n/a
Standby *3	0.04	0.16	n/a
Pin 3 pulled low	0.0001	0.0005	n/a

*1 Measured at 4 V input voltage.

*2 About 500 µs in-rush current pulse at the beginning of lasering.

*3 0.15 A during the first 2 seconds after power-up.

Table 5. Typical in-rush current after enabling power-on signal.

Input voltage [V]	In-rush current [A]	Pulse duration [ms]
3.3	2.0	2
4.0	2.0	1.5
5.0	2.0	1

3 Serial communication protocol

This chapter describes the main command frames of the laser range finder modules. Notice that measurement commands described in earlier ICD document versions are still supported.

Default baud rate is 115200 bps (no parity, 8 data bits, 1 stop bit). Serial interface is UART (3.3V). Serial transfer speed can be changed by the user.

After about 50 ms from the power on the module sends an identification string like “LRF127 x.x.x” where “x.x.x” is the firmware version (e.g., 2.5.5.4). After the string, the module is ready to take commands.

Note! Keep at least 20 milliseconds break after receiving the response frame before sending any command. Otherwise, the command may be ignored.

3.1 Standard acknowledgement frame

Standard answer does not contain any data. Notice, that Check byte is an “exclusive OR” of **50h** and **arithmetic sum** of all data including synchronization bytes.

Standard acknowledgement frame:

Byte	MSB								HEX	Note
	7	6	5	4	3	2	1	0		
-2	0	1	0	1	1	0	0	1	59h	1. synchronization byte
-1	y	y	y	y	y	y	y	y	YYh	2. synchronization byte Echo of command byte
0	0	0	1	1	1	1	0	0	3Ch	Acknowledgement byte
1	x	x	x	x	x	x	x	x	XXh	Check byte, checksum XOR 50h $XX = (59h + YYh + 3Ch) \text{ XOR } 50h$

3.2 Execute range measurement

This command executes range measurements. The measurement mode is selected by setting measurement mode byte 1. Ranging performance of each mode is presented in **Table 7**.

Table 6. Measurement mode list.

Mode	Description
SMM	Single Measurement Mode
Quick SMM 1 / 2	Quick Single Measurement Mode(s) Reduces measuring time.
CMM	Continuous Measurement Mode Used when fixed rates are needed.

Command frame

Byte	MSB				LSB				HEX	Note
	7	6	5	4	3	2	1	0		
0	1	1	0	0	1	1	0	0	CCh	Execute range measurement Measurement mode: 00h = SMM 10h = Quick SMM 1 20h = Quick SMM 2 01h = CMM 1 Hz *1 02h = CMM 4 Hz *1 03h = CMM 10 Hz *1 04h = CMM 20 Hz *1 05h = CMM 100 Hz *1 06h = CMM 200 Hz *1 07h = CMM 500 [Hz] *1, *2 Extra delay between pulses [µs] *3 Burst Divider *4 Check byte
S	0	0	X	X	X	X	X	X	XXh	
2	0	0	0	0	0	0	0	0	00h	
3	0	0	0	0	0	0	0	0	00h	
4	X	X	X	X	X	X	X	X	XXh	

*1 Module returns measurements until stopped. Continuous modes can be stopped with any command e.g. A Break command C6h 96h (see chapter 3.13).

*2 This feature is available in firmware 2.6.x.x and above

*3 Parameter can be used to reduce power consumption by slowing down the measurement. Parameter doesn't affect to the CMM modes higher than 1 Hz.

*4 Laser Pulse count can be reduced if shorter measurement time with limited range performance is required.

An example - To start 10 Hz continuous mode send the following command bytes:

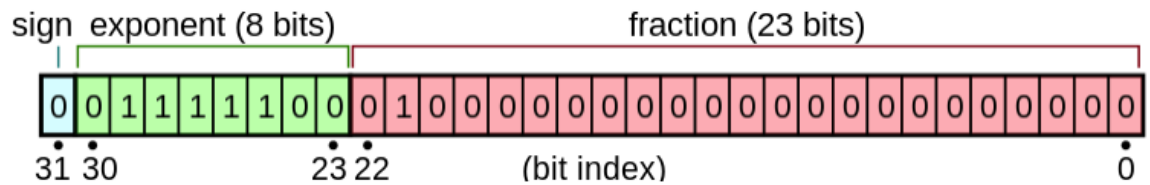
CCh 03h 00h 00h 9F

Response frame

Byte	MSB				LSB				HEX	Note
	7	6	5	4	3	2	1	0		
-2	0	1	0	1	1	0	0	1	59h	Synchronization byte
-1	1	1	0	0	1	1	0	0	CCh	Echo of command byte
0-3	X	X	X	X	X	X	X	X	XXh	Range 1 [m] *1
4	X	X	X	X	X	X	X	X	XXh	Signal level 1 (LSB)
5	X	X	X	X	X	X	X	X	XXh	Signal level 1 (MSB)
6-9	X	X	X	X	X	X	X	X	XXh	Range 2 [m] *1
10	X	X	X	X	X	X	X	X	XXh	Signal level 2 (LSB)
11	X	X	X	X	X	X	X	X	XXh	Signal level 2 (MSB)
12-15	X	X	X	X	X	X	X	X	XXh	Range 3 [m] *1
16	X	X	X	X	X	X	X	X	XXh	Signal level 3 (LSB)
17	X	X	X	X	X	X	X	X	XXh	Signal level 3 (MSB)
18	X	MT	NT	ERR	NR	TTE	X	X	XXh	Status byte #3 *2
19	X	X	X	X	X	X	X	X	XXh	Check byte (checksum XOR 50h)

*1 IEEE 754: single-precision binary floating-point format.

32-bit float, LSB first/little endian:



*2 Byte 6 is Status byte #3. Read chapter 3.4 Status query for more information.

Table 7. Typical range performance in the different measurement modes against NATO target (2.3 m x 2.3 m, albedo 30%, visibility 25 km)

Mode	LRF126/127-M4
SMM	1850
QUICK SMM 1	1500
QUICK SMM 2	1700
CMM 1 Hz	1700
CMM 4 Hz	1400
CMM 10 Hz	1200
CMM 20 Hz	1000
CMM 100 Hz	500
CMM 200 Hz	400
CMM 500 Hz	150

3.2.1 Single Measurement Mode

In the Single Measurement Mode, the LRF makes a single distance measurement with best possible performance.

Table 8. Typical measurement times and maximum ranging capabilities in SMM.

Module	Meas time [s]	Range capability [m]
LRF127-M4	1.3	4500
LRF126-M4	1.3	4500

*1 At bad weather conditions measurement time can be about 300ms longer.

The LRF returns the first, second and last targets in the merit order, the most probable target first. The user may sort the ranges to the distance order if preferred. If less than three (3) targets are found the remaining distances are zero (0.0).

Bit “Not Ready” (NR) of the status byte 1 can be polled to see when the next measurement is possible. Typical measurement rate is five (5) measurements in ten (10) seconds.

3.2.2 Quick Single Measurement Mode

A Quick Single Measurement Mode executes a single measurement with limited maximum duration.

Table 9. Typical measurement times when using QSMM

Module	QSMM 1 [s]	QSMM 2 [s]
LRF126-M4	0.45	0.8
LRF127-M4	0.45	0.8

3.2.3 Continuous Measurement Mode

When the mode is enabled, the module will start to measure at selected rate. The range data up to three different distances are returned. If no target is detected zero (0.0) value is returned as distance.

Continuous mode can be running non-stop. All modes are eye safe.

Mode will be stopped if any command is sent during continuous measurement.

3.3 Check optical crosstalk

This diagnostic command can be used to check the optical crosstalk from the front glass or mechanical structures after user integrates the LRF.

Most reliable result is achieved by aiming the LRF towards clear sky. The output indicates the maximum distance the crosstalk may have an effect to the performance. Optimal values are typically less than 100 m.

Command frame

Byte	MSB				LSB				HEX	Note
	7	6	5	4	3	2	1	0		
0	1	1	0	1	1	1	1	0	DEh	Check optical crosstalk
1	1	0	0	0	1	1	1	0	8Eh	Check byte

Response frame

Byte	MSB				LSB				HEX	Note
	7	6	5	4	3	2	1	0		
-2	0	1	0	1	1	0	0	1	59h	Synchronization byte
-1	1	1	0	1	1	1	1	0	DEh	Echo of command byte
0	x	x	x	x	x	x	x	x	XXh	Effect range (LSB) [m]
1	x	x	x	x	x	x	x	x	XXh	Effect range (MSB)
2	x	x	x	x	x	x	x	x	XXh	Check byte

3.4 Status query

This command returns the status bytes of the LRF.

Command frame

Byte	MSB				LSB				HEX	Note
	7	6	5	4	3	2	1	0		
0	1	1	0	0	0	1	1	1	C7h	Ask Status
1	1	0	0	1	0	1	1	1	97h	Check byte

Response frame

Status byte 3 is the same as in command CCh (see Chapter 3.2).

Byte	MSB				LSB				HEX	Note
	7	6	5	4	3	2	1	0		
-2	0	1	0	1	1	0	0	1	59h	Synchronization byte
-1	1	1	0	0	0	1	1	1	C7h	Echo of command byte
0	GP	TP	REB	NR	-	-	RP	LP	XXh	Status byte #1 *1
1	PNTR	HV	-	DC	MEM	-	LB	CP	XXh	Status byte #2 *1
2	-	MT	NT	ERR	x	TTE	-	-	XXh	Status byte #3 *1
3	x	x	x	x	x	x	x	x	XXh	Check byte

***1** Status bytes #1, #2 and #3 of status query response frame:

Name	Byte.bit	Meaning, status byte 1	Action
GP	0.7	General problems	Contact service
TP	0.6	Transmitter problems	Contact service
REB	0.5	Rebooted since last status query	
NR	0.4	Not Ready to perform. Too high requested measurement frequency	Measurement is not possible before the bit clears
	0.3	Not used	
	0.2	Not used	
RP	0.1	Receiver problems	Contact service *2
LP	0.0	Laser power problems	Contact service *2
Name	Byte.bit	Meaning, status byte 2	Action
VPOINT	1.7	Visual pointer ON	
HV	1.6	High voltage out of limits	
	1.5	Not used	
DC	1.4	DCDC output out of limits	Contact service *2
MEM	1.3	Memory problem	Contact service *2
	1.2	Not used	
LB	1.1	Low battery	Check power supply
CP	1.0	Communication problem, last received	Check communication link
Name	Byte.bit	Meaning, status byte 3	Action
	2.7/6.7	Not used	
MT	2.6/6.6	Multiple Targets	
NT	2.5/6.5	No Targets	
ERR	2.4/6.4	Error reported by LRF. More information in status bytes 1 & 2	Contact service or use command C7h to check status bytes 1 & 2
	2.3/6.3	Not used	
TTE	2.2/6.2	Transmitter timing error	Contact service
	2.1/6.1	Not used	
	2.0/6.0	Not used	

***2** If module operates normally, it is safe to use until it can be shipped to be serviced.

3.5 Set alignment pointer

This command controls the alignment pointer. The pointer is switched off automatically after range finding. Note: Alignment pointer is available in LRF127 device only.

Command frame

MSB				LSB						Note
Byte	7	6	5	4	3	2	1	0	HEX	
0	1	1	0	0	0	1	0	1	C5h	Set pointer mode 0 – pointer OFF 1 – reserved 2 – visible pointer ON 3 – reserved Check byte (checksum XOR 50h)
1	0	0	0	0	0	0	x	x	0Xh	
2	x	x	x	x	x	x	x	x	XXh	

Answer: Standard acknowledgement frame.

3.6 Ask range window

This command returns the range window setting in use. Range window is used to select the operational distance range.

Command frame

MSB					LSB						
Byte	7	6	5	4	3	2	1	0	HEX	Note	
0	0	0	1	1	0	0	0	0	30h	Ask Range Window	
1	0	1	1	0	0	0	0	0	60h	Check byte	

Response frame

	MSB				LSB					
Byte	7	6	5	4	3	2	1	0	HEX	Note
-2	0	1	0	1	1	0	0	1	59h	Synchronization byte
-1	0	0	1	1	0	0	0	0	30h	Echo of command byte
0	x	x	x	x	x	x	x	x	XXh	Minimum Range (LSB) [m]
1	x	x	x	x	x	x	x	x	XXh	Minimum Range (MSB)
2	x	x	x	x	x	x	x	x	XXh	Maximum Range (LSB) [m]
3	x	x	x	x	x	x	x	x	XXh	Maximum Range (MSB)
4	x	x	x	x	x	x	x	x	XXh	Check byte (checksum XOR 50h)

3.7 Set minimum range

Ignore targets closer than minimum range setting. Maximum accepted value is five (5) meters less than Max Range setting. Minimum value is zero (0) meters.

Note: Actual achievable minimum range is affected by possible front glass or rain/fog/snow reflections. To save range window setting into the permanent memory see chapter 3.9.

Command frame

Byte	MSB				LSB				HEX
	7	6	5	4	3	2	1	0	
0	0	0	1	1	0	0	0	1	31h
1	x	x	x	x	x	x	x	x	XXh
2	x	x	x	x	x	x	x	x	XXh
3	x	x	x	x	x	x	x	x	XXh

Note
Set minimum range
Min Range (LSB) [m]
Min Range (MSB)
Check byte (checksum XOR 50h)

Answer: Standard acknowledgement frame.

3.8 Set maximum range

Ignore targets further than the maximum range setting. Minimum accepted value is 5 meters more than Minimum Range setting.

Note: To save range window setting into the permanent memory see chapter 3.9.

Command frame

Byte	MSB				LSB				HEX
	7	6	5	4	3	2	1	0	
0	0	0	1	1	0	0	1	0	32h
1	x	x	x	x	x	x	x	x	XXh
2	x	x	x	x	x	x	x	x	XXh
3	x	x	x	x	x	x	x	x	XXh

Note
Set maximum range
Max Range (LSB) [m]
Max Range (MSB)
Check byte (checksum XOR 50h)

Answer: Standard acknowledgement frame.

3.9 Set baud rate or save settings

This command sets baud rate of the module. Default baud rate is 115200 bps. The command changes baud rate immediately after acknowledgement frame. The setting can be saved to permanent memory by using baud rate selection zero (0). The current range window setting is also saved to permanent memory.

For example: 38400 bps is set by sending bytes: C8h 03h 9Bh.

Command frame

Byte	MSB				LSB				HEX
	7	6	5	4	3	2	1	0	
0	1	1	0	0	1	0	0	0	C8h
1	0	0	0	0	0	x	x	x	0Xh
2	1	0	0	1	1	x	x	x	9Xh

Note
Set baud rate or save settings
Baud rate selection: 0 = Save setting 1 = 9600 bps (not recommended) 2 = 19200 bps 3 = 38400 bps 4 = 57600 bps 5 = 115200 bps (default) 6 = 230400 bps 7 = 460800 bps
Check byte (checksum XOR 50h)

Answer: Standard acknowledgement frame.

3.10 Request identification

This command requests identification information from the LRF.

Command frame

MSB					LSB				
Byte	7	6	5	4	3	2	1	0	HEX
0	1	1	0	0	0	0	0	0	C0h
1	1	0	0	1	0	0	0	0	90h

Note
Request identification info
Check byte

Response frame

Byte	MSB				LSB				HEX	Note
	7	6	5	4	3	2	1	0		
-2	0	1	0	1	1	0	0	1	59h	Synchronization byte
-1	1	1	0	0	0	0	0	0	C0h	Echo of command byte
0-14	x	x	x	x	x	x	x	x	XXh	15 bytes: LRF ID string
15	0	0	0	0	1	1	0	1	0Dh	CR (Carriage Return)
16	0	0	0	0	1	0	1	0	0Ah	LF (Line Feed)
17-31	x	x	x	x	x	x	x	x	XXh	15 bytes Reserved for additional information
32	0	0	0	0	1	1	0	1	0Dh	CR
33	0	0	0	0	1	0	1	0	0Ah	LF
34-43	x	x	x	x	x	x	x	x	XXh	10 bytes: Serial number string
44	0	0	0	0	1	1	0	1	0Dh	CR
45	0	0	0	0	1	0	1	0	0Ah	LF
46	x	x	x	x	x	x	x	x	XXh	Firmware version (LSB)
47	0	0	0	0	0	0	0	0	XXh	Firmware version (MSB)
48	1	0	1	1	0	0	0	1	B1h	Electronics type
49	1	0	1	1	0	0	0	0	B0h	Optics type
50-57	x	x	x	x	x	x	x	x	XXh	8 bytes: Date YY-MM-DD *1
58	0	0	0	0	1	1	0	1	0Dh	CR
59	0	0	0	0	1	0	1	0	0Ah	LF
60-67	x	x	x	x	x	x	x	x	XXh	8 bytes: Time HH:MM:SS *1
68	0	0	0	0	1	1	0	1	0Dh	CR
69	0	0	0	0	1	0	1	0	0Ah	LF
70	x	x	x	x	x	x	x	x	XXh	Check byte (checksum XOR 50h)

*1 Date and time of the firmware

3.11 Request diagnostic data

Information frame includes data which can be used in diagnostic purposes.

Command frame

Command name

Byte	MSB				LSB				HEX	Note
	7	6	5	4	3	2	1	0		
0	1	1	0	0	0	0	1	0	C2h	Request diagnostic data
1	1	0	0	1	0	0	1	0	92h	Check byte

Response frame

Byte	MSB				LSB				HEX	Note
	7	6	5	4	3	2	1	0		
-2	0	1	0	1	1	0	0	1	59h	Synchronization byte
-1	1	1	0	0	0	0	1	0	C2h	Echo of command byte
0	x	x	x	x	x	x	x	x	XXh	Reserved diagnostic data
1									XXh	Reserved diagnostic data
2									XXh	Reserved diagnostic data
3									XXh	Reserved diagnostic data
4									XXh	Reserved diagnostic data
5									XXh	Reserved diagnostic data
6									XXh	Reserved diagnostic data
7									XXh	Reserved diagnostic data
8									XXh	Target 1 distance [1 m], LSB
9									XXh	Target 1 distance, MSB
10									XXh	Target 2 distance [1 m], LSB
11									XXh	Target 2 distance, MSB
12									XXh	Target 3 distance [1 m], LSB
13									XXh	Target 3 distance, MSB
14									XXh	Target 1 magnitude
15									XXh	Target 2 magnitude
16									XXh	Target 3 magnitude
17									XXh	Not in use
18									XXh	Battery voltage (+U [mV]), LSB *1
19									XXh	Battery voltage, MSB *1
20									XXh	Not used
21									XXh	Not used
22									XXh	IO voltage (+3300 mV), LSB
23									XXh	IO voltage, MSB
24									XXh	RX voltage ([0.01V]), LSB
25									XXh	RX voltage, MSB
26									XXh	TX voltage [mV], LSB
27									XXh	TX voltage [mV], MSB
28									XXh	RX temperature [0.01°C], LSB
29									XXh	RX temperature [0.01°C], MSB
30									XXh	Status byte 1 *2
31									XXh	Status byte 2 *2
32									XXh	Status byte 3 *2
33									XXh	Pulse Counter [millions of pulses] (LSB)
34									XXh	Pulse Counter
35									XXh	Pulse Counter (MSB)
36									XXh	Serial error counter
37	x	x	x	x	x	x	x	x	XXh	Check byte (checksum XOR 50h)

*1 Battery voltage is not updated at CMM 10 Hz and faster modes.

*2 For more information about status bytes check chapter 3.4.

3.12 Reset serial error counter

Error counter is returned in the diagnostic data (see chapter 3.11 Request diagnostic data).

Command frame

Byte	MSB				LSB				HEX	Note
	7	6	5	4	3	2	1	0		
0	1	1	0	0	1	0	1	1	CBh	Reset error counter of serial communication
1	1	0	0	1	1	0	1	1	9Bh	Check byte (checksum XOR 50h)

Answer: Standard acknowledgement frame.

3.13 Break range measurement

The command can be used to stop continuous measurement.

Command frame

Byte	MSB				LSB				HEX	Note
	7	6	5	4	3	2	1	0		
0	1	1	0	0	0	1	1	0	C6h	Break range measurement
1	1	0	0	1	0	1	1	0	96h	Check byte (checksum XOR 50h)

Answer: Standard acknowledgement frame.